Thyroid Function Review

CASE 01 | FICTIONAL PERSON | NO REAL PATIENT DATA

Synthetic model-evaluation document. Not for clinical use. Do not use these values to make decisions about a real person.

Referral note

A 42-year-old nonpregnant woman reports four months of fatigue, constipation, cold sensitivity, and gradual weight gain. Symptoms began before the current laboratory evaluation. She denies neck pain, recent iodinated contrast, and acute illness.

Examination note: pulse 58 beats/min, temperature 36.4 C. Thyroid is mildly and diffusely enlarged without a discrete nodule. No tremor or eye findings are recorded.

Medical history	No pituitary disease, thyroid surgery, radioiodine exposure, or prior thyroid diagnosis.
Medication and supplements	No thyroid medication, lithium, amiodarone, or high-dose biotin.
Family history	One first-degree relative has autoimmune thyroid disease.
Collection	Outpatient blood draw; not pregnant; no acute infection reported.

All names, dates, histories, values, intervals, and notes in this case are fabricated.

Initial laboratory report

Results from the same serum sample are shown below. The laboratory flags results against its printed adult reference intervals.

Test	Result	Unit	Printed reference interval	Report flag
TSH	18.6	mIU/L	0.40 - 4.00	H
Free T4	0.52	ng/dL	0.80 - 1.80	L
Total T3	78	ng/dL	76 - 181	
TPO antibody	684	IU/mL	< 35	H
Hemoglobin	13.1	g/dL	12.0 - 16.0	
Sodium	139	mmol/L	135 - 145	

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Repeat and context

A second sample was collected six weeks later while the patient remained clinically stable. No thyroid-directed medication was started between samples.

Test	Result	Unit	Printed reference interval	Report flag
TSH, repeat	17.9	mIU/L	0.40 - 4.00	H
Free T4, repeat	0.55	ng/dL	0.80 - 1.80	L
TPO antibody	Not repeated			
Pregnancy test	Negative			

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